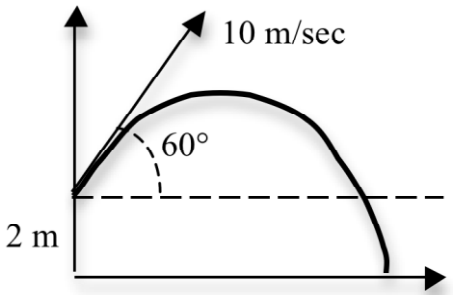
BC 3 Name: _____

Vectors 3: Projectile Motion

Example:

A person throws a ball into the air from a height of 2 meters. The initial speed of the ball is 10 meters/second and the angle of elevation is 60°.



$\vec{r}(0) = 2\vec{j}$

initial speed $= \left|\vec{v}(0)\right| = 10$ m/sec

$\vec{v}(0) = \left(10\cos(60°)\right)\vec{i} + \left(10\sin(60°)\right)\vec{j}$

$= 5\vec{i} + 5\sqrt{3}\,\vec{j}$

We know that the acceleration is $\vec{a}(t) = -9.8\,\vec{j}$, due to gravity. From this, we have

$$\vec{v}(t) = -9.8t\,\vec{j} + \vec{v}(0)$$
$$= 5\vec{i} + \left(-9.8t + 5\sqrt{3}\right)\vec{j}$$

This allows us to find $\vec{r}(t)$.

$$\vec{r}(t) = 5t\,\vec{i} + \left(-4.9t^2 + 5\sqrt{3}\,t\right)\vec{j} + \vec{r}(0)$$
$$= 5t\,\vec{i} + \left(-4.9t^2 + 5\sqrt{3}\,t + 2\right)\vec{j}$$

The corresponding parametric equations are:

$$x(t) = 5t \text{ and } y(t) = -4.9t^2 + 5\sqrt{3}\,t + 2 \text{ for } t \geq 0$$

What is the maximum height reached by the ball?

Occurs when $\dfrac{dy}{dt} = -9.8t + 5\sqrt{3} = 0 \Rightarrow t = 0.884\,\text{sec} \Rightarrow y = 5.827\,\text{m}$

How far (horizontally) will the ball travel before it hits the ground?

Occurs when $y(t) = 0 \Rightarrow -4.9t^2 + 5\sqrt{3}\,t + 2 = 0 \Rightarrow t = 1.974\,\text{sec} \Rightarrow x(1.974) = 9.871\,\text{m}$

What will be the speed of the ball at the instant it hits the ground?

$$\text{speed} = \left|\vec{v}(1.974)\right| = \sqrt{\left(x'(1.974)\right)^2 + \left(y'(1.974)\right)^2} = \sqrt{5^2 + (-10.686)^2} \approx 11.798\,\text{m/sec}$$

Please complete the following problems on a separate piece of paper.

1. A baseball is thrown from the stands 32 ft above the field at an angle of 36°. When and how far away will the ball strike the ground if its initial speed is 32 ft/sec?

2. What two angles of elevation will enable a projectile to reach a target 16 km away at the same elevation as the gun that fired the projectile, if the projectile's initial speed is 400 m/sec?

3. An electron in a TV tube is beamed horizontally at a speed of 5×10^6 m/sec toward the face of the tube 40 cm away. About how far will the electron drop before it hits the tube?

4. Laboratory tests designed to determine how far golf balls of different hardness travel when hit with a driver showed that a certain golf ball hit with a club-head speed of 100 mph at a launch angle of 9° carried 248.8 yards. What was the launch speed of the ball (and no, it was NOT 100 mph)?

5. Show that a projectile attains three-quarters of its maximum height in half the time it takes to reach its maximum height.

6. A golf ball leaves the ground at a 30° angle at a speed of 90 ft/sec. Will it clear the top of a 30-ft tree 135 ft away?